RAJALAKSHMI ENGINEERING COLLEGE

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B2

Started on	Saturday, 12 September 2026, 1:54 PM
State	Finished
Completed on	Saturday, 12 September 2026, 1:54 PM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[1000], b[1000];
    int dp[1000][1000];

    scanf("%s", a);
    scanf("%s", b);

    int n = strlen(a), m = strlen(b);

    for (int i = 0; i <= n; i++) {
        for (int j = 0; j <= m; j++) {
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (a[i-1] == b[j-1])
                dp[i][j] = dp[i-1][j-1] + 1;
            else
                dp[i][j] = dp[i-1][j] > dp[i][j-1] ? dp[i-1]
        }
    }

    printf("%d", dp[n][m]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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